Junehyoung Jeon
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Research Interests

I am interested in how the early Universe and its components such as galaxies, first stars, and black holes evolved. I approach these problems creating/using theoretical models and hydrodynamical simulations.

Education

Doctor of Philosophy in Astronomy — Expected completion May 2027
Department of Astronomy, University of Texas at Austin, Austin, Texas

Master of Arts in Astronomy — August 2023
Department of Astronomy, University of Texas at Austin, Austin, Texas

Bachelor of Science in Astrophysics, Minor in Physics — May 2021
Barrett Honors College, Arizona State University, Tempe, Arizona
Average unweighted GPA: 4.0/4.0

Publications

First-Author Publications

Jeon, J. et al. (2026). *Heavy Seeds and the Multi-scale Fueling of Early Supermassive Black Holes and Little Red Dots.* arXiv:2610.36089, Submitted to The Astrophysical Journal.

Jeon, J. et al. (2026). *Early Supermassive Black Holes and Little Red Dots Require Free-Fall Growth.* arXiv:2609.00115, Submitted to the Open Journal of Astrophysics.

Jeon, J. et al. (2026). *What is Powering the Enigmatic He* II *Emitter Hebe: The First Stars or Black Holes?.* The Astrophysical Journal, Volume **1006**, Issue 1, 27.

Jeon, J. et al. (2026). *Hunting for the First Explosions at the High-Redshift Frontier.* The Astrophysical Journal, Volume **1001**, Issue 1, 3.

Jeon, J. et al. (2026). *Little Red Dots and their Progenitors from Direct Collapse Black Holes.* The Astrophysical Journal, Volume **998**, Issue 1, 148.

Jeon, J. et al. (2025). *The Emerging Black Hole Mass Function in the High-Redshift Universe.* The Astrophysical Journal, Volume **988**, Issue 1, 110.

Jeon, J. et al. (2025). *Physical Pathways for JWST-Observed Supermassive Black Holes in the Early Universe.* The Astrophysical Journal, Volume **979**, Issue 2, 127.

Jeon, J. et al. (2023). *Observability of Low-Luminosity AGN in the Early Universe with JWST.* Monthly Notices of the Royal Astronomical Society, **524**, 176-187.

Jeon, J. et al. (2022). *Maximal X-ray feedback in the pre-reionization Universe.* Monthly Notices of the Royal Astronomical Society, **515**, 5568-5575.

Co-Author Publications

Taylor, A., et al. (**Incl. Jeon**) (2026) . *Little Red Dots with Broad Hβ Lines Reveal the Black*

Hole Mass Function at $7 < z < 9.5$. arxiv:2610.37564, Submitted to The Astrophysical Journal.

Zhang, S., **Jeon, J.**, et al. (2026). *The Road to Normalcy: Environment-Driven Evolutionary Pathways for Primordial Black Holes.* arxiv:2608.23498, Submitted to The Astrophysical Journal.

Gupta, A., et al. (**Incl. Jeon**) (2026). *A Rapid Evolution in the Observed* $M_{\rm BH}/M_*$ *Relation at* $z > 3$ *Revealed via Spectro-photometric SED-Modeling.* arxiv:2605.30414, Submitted to The Astrophysical Journal.

Fei, Q., et al. (**Incl. Jeon**) (2026). *A GLIMPSE of Intermediate Mass Black holes in the epoch of reionization: Witnessing the Descendants of Direct Collapse?.* The Astrophysical Journal, Volume **1003**, Issue 2, 244.

Zhang, S., Liu, B., Bromm, V., **Jeon, J.**, et al. (2025). *How do Massive Primordial Black Holes Impact the Formation of the First Stars and Galaxies?.* The Astrophysical Journal, Volume **987**, Issue 2, 185.

Taylor, A., Finkelstein, S., Kocevski, D., **Jeon, J.**, et al. (2025). *Broad-Line AGN at* $3.5 < z < 6$*: The Black Hole Mass Function and a Connection with Little Red Dots.* The Astrophysical Journal, Volume **986**, Issue 2, 165.

Invited Talks

Heavy Seeds and the Multi-scale Fueling of Early Supermassive Black Holes and Little Red Dots
Theoretical Astroparticle and Cosmology Symposium, Texas A&M University — October 2026

Conference and Seminar Talks

The Emerging Black Hole Mass Function in the High-Redshift Universe
Shifting Landscapes in Astrophysics: New Frontiers to Explore with Roman, Pasadena, United States — July 2026
Cosmic Frontier Center Inaugural Conference, Austin, United States — May 2025

Little Red Dots and their Progenitors from Direct Collapse Black Holes
Little Red Dots 2026, Online Workshop — June 2026
ITC Lunch, Harvard Center for Astrophysics — September 2025

Hunting for the First Explosions at the High-Redshift Frontier
Spring Symposium, Space Telescope Science Institute — May 2026

Formation and Evolution of Supermassive Black Holes in the Early Universe
Galaxy Formation Seminar, Center for Computational Astrophysics — October 2025
Seminar, Columbia University Astronomy & Astrophysics — October 2025
MAT talks, MIT Kavli Institute — September 2025
HEAD Seminar, Harvard Center for Astrophysics — September 2025
Galaxy and AGN Journal Club, Space Telescope Science Institute — September 2025

Physical Pathways for JWST-Observed Supermassive Black Holes in the Early Universe
CRISOL25, Toledo, Spain (remote) — May 2025
Probing the Genesis of SMBH, Kashiwa, Japan (remote) — November 2024
CCA-CFC Workshop, Austin, United States — November 2024
ITC Lunch, Harvard Center for Astrophysics — November 2024

Conditions for Efficient Growth of Supermassive Black Holes in the Early Universe
COSPAR 2024, Busan, South Korea — July 2024
Massive Black Holes in the First Billion Years, Kinsale, Ireland — April 2024

Observability of Low-Luminosity AGN in the Early Universe with JWST
Black Holes on Broadway, New York City, United States — December 2023
Young Astronomers on Galactic Nuclei, Palermo, Italy — October 2023

Awards and Honors

Research covered by AAS Nova and Astrobites
College Continuing Graduate Fellowship, UT Austin ($10,000) — 2026
David Alan Benfield Memorial Fellowship in Astronomy, UT Austin ($1,500) — 2025
Graduate School Dissertation Writing Fellowship, UT Austin ($24,000) — 2025
Graduate Summer Excellence Award, UT Austin ($10,000) — 2025
Board of Visitors Graduate Student Endowment Fund ($10,000), UT Austin — 2023
Professional Development Award, UT Austin ($500) — 2023
The College of Liberal Arts and Sciences Dean's Medal, ASU — 2021
Moeur Award, ASU Alumni Association — 2021
New American University Award ($5,000) — 2018–2021
The College of Liberal Arts and Sciences Dean's List, ASU — 2018–2021

Professional Services

Referee for Physical Review Letters, Physical Review D, and the Astrophysical Journal

Teaching and Mentoring

Teaching Assistant, University of Texas at Austin

- *Astrophysics* — Fall 2026, Spring 2025
  - Assisted astronomy and physics major upper class students with assignments and projects, helping their understanding of the lecture concepts and applications.
  - Prepared and taught introductory python modules designed for students who have no prior experience, introducing them to the basics of coding in astrophysics.
- *Practical Introduction to Research* — Spring 2024, Fall 2021
  - Guided astronomy and physics major students through in-class activities such as coding, writing resume, and making posters, providing them with introductory skills in academia.
  - Prepared coding and lecture modules for the students, leading the students to understand the concepts.
- *Cosmology* — Fall 2022
  - Explained course topics to students after class, helping them individually to grasp lecture concepts that they might not have fully understood.
  - Guided students through their homeworks, assisting them in understanding questions and steps to solve various cosmology problems.

Undergraduate Research Projects, University of Texas at Austin — Fall 2024 – Current

- Mentored undergraduate students on research projects on the first supermassive black holes, enabling the students to perform theoretical modeling and analysis.
- Guided the students in writing research publications and conference presentations, producing concrete results from the students' work.

Mentor, Undergraduate Mentorship in Astronomy Program,
University of Texas at Austin Fall 2023 – Current

- Advised multiple undergraduate mentees on various topics, including research, graduate school preparation, and applications, helping them advance their academic career.
- Held weekly meetings to discuss updates and new questions brought up by the mentees, guiding them in general academics.

Skills

Programming: Python, C, Fortran, High Performance Computing
Operating systems: Windows, Linux
Software: LaTeX, Mathematica
Languages: English, Korean